

Domain and range and set notation



1

a $\{1, 2, 3, 4\}$

b $\{-7, -6, -5, -4\}$

c $\{1, 2, 3, 6, 7, 14, 21, 42\}$

2

a $\in$

b $\notin$

c $\in$

d $\notin$

e $\notin$

3

a False

b True

c True

d True

e False

f True

g True

4 Example answers:

$\{x | x \text{ is an even number and } 2 \leq x \leq 8\}$

$\{x | x \text{ is a positive even number and } x < 10\}$

5

a
i $\{8, 1, 3, 9, 2\}$ ii $\{5, 3, 4, 1, 9\}$ iii Yes

b
i $\{2, 3, 6, 8\}$ ii $\{3, 8, 1, 7, 2\}$ iii No

c
i $\{4, 8, 2, 7, 3\}$ ii $\{4, 8, 9, 1\}$ iii Yes

d
i $\{1, 6, 3, 8, 2\}$ ii $\{3, 2, 7, 1\}$ iii Yes

e
i $\{7, 8, 5, 3\}$ ii $\{1, 9, 3, 2, 6\}$ iii No

f
i $\{2, 6, 9, 1, 5\}$ ii $\{6, 5, 1, 8, 3\}$ iii Yes

6

a
i $\{2, 4, 6, 8\}$ ii $\{2, 4, 6, 8, 10\}$ iii No

b
i $3 \leq x \leq 7$ ii $5 \leq y \leq 8$ iii Yes

c
.....

d

i $x = 5$

ii $2 \leq y \leq 6$



iii No

e

i $-9 \leq x \leq 9$

ii $-9 \leq y \leq 9$

iii No

f

i $-7 \leq x \leq 7$

ii $0 \leq y \leq 7$

iii Yes

g

i All real x

ii $y \geq 0$

iii Yes

h

i $x \geq 0$

ii All real y

iii No

7 $-6 \leq x \leq 6$

8

a

i All real values

ii $\{y|y \leq 5\}$

b

i $\{x|x < 5\}$

ii $\{y|y > -2\}$

c

i All real values

ii $\{y|y \geq 0\}$

9

a No

b No

c Yes

d No

10

a No

b No

c No

d Yes

11

a No

b Yes

c No

d No

12 $\left[-\frac{1}{2}, \infty\right)$

13

a $\{y|y \leq 125\}$

b $\{y|0 \leq y \leq 125\}$

14



15

a Yes

b No

c No

d No

Additional questions

16

a $\{1, 2, 3, 4\}$

b $\{1, 2, 3, 4\}$

17

a $\{1, 3, 6, 9\}$

b $\{2, 5, 7, 9\}$

18

a $\{-1, 3, 16, 19\}$

b $\{2, 5, 7\}$

19

a $\{-2, 0, 2, 4\}$

b $\{12, 51, 70\}$

20

a Possible answers:

- Domain: $-4 \leq x \leq 7$, Range: $-5 \leq y \leq 7$
- Domain: $[-4, 7]$, Range: $[-5, 7]$
- Domain: $\{x | -4 \leq x \leq 7\}$, Range: $\{y | -5 \leq y \leq 7\}$

b Possible answers:

- Domain: $-2 < x \leq 4$, Range: $-6 \leq y < 5$
- Domain: $(-2, 4]$, Range: $[-6, 5)$
- Domain: $\{x | -2 < x \leq 4\}$, Range: $\{y | -6 \leq y < 5\}$

c Possible answers:

- Domain: $-4 < x < 6$, Range: $-5 < y < 3$
- Domain: $(-4, 6)$, Range: $(-5, 3)$
- Domain: $\{x | -4 < x < 6\}$, Range: $\{y | -5 < y < 3\}$

d Possible answers:



- Domain: $-1 \leq x \leq 3$, Range: $-1 \leq y \leq 6$
- Domain: $[-1, 3]$, Range: $[-1, 6]$
- Domain: $\{x | -1 \leq x \leq 3\}$, Range: $\{y | -1 \leq y \leq 6\}$

e Possible answers:

- Domain: $(-\infty, \infty)$, Range: $(-\infty, \infty)$
- Domain: All real values, Range: All real values

f Domain: All integer values, Range: All integer values

g Possible answers:

- Domain: $x \geq -5$, Range: All real values
- Domain: $[-5, \infty]$, Range: $[-\infty, \infty]$
- Domain: $\{x | x \geq -5\}$, Range: All real values

h Possible answers:

- Domain: $x = 4$, Range: $-6 < y < 6$
- Domain: $x = 4$, Range: $(-6, 6)$
- Domain: $\{x | x = 4\}$, Range: $\{y | -6 < y < 6\}$

21

a The domain is only $x \leq 8$

b The domain consists of all the real numbers

22

a The range consists of all the real numbers

b The range is only $y = 2$

23

a Domain: $35 \leq x \leq 50$
Range: $200 \leq y \leq 450$

b Domain: $0 \leq x \leq 17$
Range: $112 \leq y \leq 237$

c Domain: $0 \leq x \leq 134$
Range: $0 \leq y \leq 26.2$

24

a $x \geq 0$

b $y \geq 0.14$

25



- a 40 beats per minute
- b 105°F
- c The function output $f(10)$ is the average resting heart rate of a person at 10°F according to the model. However, since $T = 10$ falls well outside of the domain of the function, we do not have enough information to consider this a reliable estimate.